

import libraries

```
In [2]: import numpy as np
import pandas as pd
```

Creating DataFrame

```
In [3]: ## using lists
student_data = [
    [100,80,10],
    [90,70,7],
    [120,100,14],
    [80,50,2]
]

pd.DataFrame(student_data,columns=['iq','marks','package'])
```

Out[3]:

	iq	marks	package
0	100	80	10
1	90	70	7
2	120	100	14
3	80	50	2

```
In [4]: # using dicts

student_dict = {
    'name':['marina','manisha','eshu','sangeetha','sneha','campusx'],
    'iq':[100,90,120,80,0,0],
    'marks':[80,70,100,50,0,0],
    'package':[10,7,14,2,0,0]
}

students = pd.DataFrame(student_dict)
students.set_index('name',inplace=True)
students
```

Out[4]:

	iq	marks	package
name			
marina	100	80	10
manisha	90	70	7
eshu	120	100	14
sangeetha	80	50	2
sneha	0	0	0
campusx	0	0	0

In []:

```
In [5]: # using read_csv
movies = pd.read_csv('movies.csv')
movies.head()
```

Out[5]:

	title_x	imdb_id	poster_path	
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
1	Battalion 609	tt9472208	NaN	https://
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.
4	Evening Shadows	tt6028796	NaN	https://en.v

```
In [6]: ipl = pd.read_csv('ipl-matches.csv')
ipl.head()
```

Out [6]:

	ID	City	Date	Season	MatchNumber	Team1	Team2	
0	1312200	Ahmedabad	2022-05-29	2022	Final	Rajasthan Royals	Gujarat Titans	Ahr
1	1312199	Ahmedabad	2022-05-27	2022	Qualifier 2	Royal Challengers Bangalore	Rajasthan Royals	Ahr
2	1312198	Kolkata	2022-05-25	2022	Eliminator	Royal Challengers Bangalore	Lucknow Super Giants	
3	1312197	Kolkata	2022-05-24	2022	Qualifier 1	Rajasthan Royals	Gujarat Titans	
4	1304116	Mumbai	2022-05-22	2022	70	Sunrisers Hyderabad	Punjab Kings	W

Selecting columns from a DataFrame

```
In [7]: #single columns
movies['original_title']
```

```
Out[7]: 0          Uri: The Surgical Strike
1          Battalion 609
2    The Accidental Prime Minister
3          Why Cheat India
4          Evening Shadows
...
1624    Tera Mera Saath Rahen
1625    Yeh Zindagi Ka Safar
1626    Sabse Bada Sukh
1627          Daaka
1628    Humsafar
Name: original_title, Length: 1629, dtype: object
```

```
In [8]: ipl['Umpire1']
```

```
Out[8]: 0      CB Gaffaney
1      CB Gaffaney
2      J Madanagopal
3      BNJ Oxenford
4      AK Chaudhary
...
945     BF Bowden
946     SJ Davis
947     Aleem Dar
948     MR Benson
949     Asad Rauf
Name: Umpire1, Length: 950, dtype: object
```

```
In [9]: #multiple column
ipl[['Team1', 'Team2', 'WinningTeam']]
```

```
Out[9]:
```

	Team1	Team2	WinningTeam
0	Rajasthan Royals	Gujarat Titans	Gujarat Titans
1	Royal Challengers Bangalore	Rajasthan Royals	Rajasthan Royals
2	Royal Challengers Bangalore	Lucknow Super Giants	Royal Challengers Bangalore
3	Rajasthan Royals	Gujarat Titans	Gujarat Titans
4	Sunrisers Hyderabad	Punjab Kings	Punjab Kings
...	...	...	...
945	Kolkata Knight Riders	Deccan Chargers	Kolkata Knight Riders
946	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore
947	Delhi Daredevils	Rajasthan Royals	Delhi Daredevils
948	Kings XI Punjab	Chennai Super Kings	Chennai Super Kings
949	Royal Challengers Bangalore	Kolkata Knight Riders	Kolkata Knight Riders

950 rows × 3 columns

```
In [10]: m1 = movies[['original_title', 'summary', 'imdb_rating']]
```

```
In [11]: m1.sort_values(ascending = False, by = 'imdb_rating').head(3)
```

```
Out[11]:
```

	original_title	summary	imdb_rating
41	Family of Thakurganj	The film is based on small town of North India...	9.4
1628	Humsafar	Ashar and Khirad are forced to get married due...	9.0
62	Saand Ki Aankh	Both over the age of 60 Chandro Tomar and Pra...	8.9

```
In [1]: # Selecting rows from a DataFrame
# ^^ iloc - searches using index positions
# ^^ loc - searches using index labels
```

```
In [13]: movies.iloc[4]
```

```
Out[13]: title_x                    Evening Shadows
imdb_id                    tt6028796
poster_path                NaN
wiki_link                  https://en.wikipedia.org/wiki/Evening_Shadows
title_y                    Evening Shadows
original_title              Evening Shadows
is_adult                    0
year_of_release             2018
runtime                     102
genres                      Drama
imdb_rating                  7.3
imdb_votes                  280
story                      While gay rights and marriage equality has bee...
summary                    Under the 'Evening Shadows' truth often plays...
tagline                     NaN
actors                     Mona Ambegaonkar|Ananth Narayan Mahadevan|Deva...
wins_nominations            17 wins & 1 nomination
release_date                11 January 2019 (India)
Name: 4, dtype: object
```

```
In [14]: ipl.iloc[945]
```

```
Out[14]: ID                    335986
City                    Kolkata
Date                    2008-04-20
Season                  2007/08
MatchNumber              4
Team1                    Kolkata Knight Riders
Team2                    Deccan Chargers
Venue                    Eden Gardens
TossWinner               Deccan Chargers
TossDecision              bat
SuperOver                 N
WinningTeam              Kolkata Knight Riders
WonBy                     Wickets
Margin                    5.0
method                   NaN
Player_of_Match          DJ Hussey
Team1Players              ['WP Saha', 'BB McCullum', 'RT Ponting', 'SC G...
Team2Players              ['AC Gilchrist', 'Y Venugopal Rao', 'VVS Laxma...
Umpire1                   BF Bowden
Umpire2                   K Hariharan
Name: 945, dtype: object
```

```
In [15]: students
```

Out[15]:

	iq	marks	package
name			
marina	100	80	10
manisha	90	70	7
eshu	120	100	14
sangeetha	80	50	2
sneha	0	0	0
campusx	0	0	0

In [16]: `students.loc['eshu']`

Out[16]:

iq	120
marks	100
package	14

Name: eshu, dtype: int64

In [17]: `students.loc['marina':'sangeetha']`

Out[17]:

	iq	marks	package
name			
marina	100	80	10
manisha	90	70	7
eshu	120	100	14
sangeetha	80	50	2

In [18]: `movies.iloc[:4]`

Out[18]:		title_x	imdb_id	poster_path
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
1	Battalion 609	tt9472208		NaN https://
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.

In [19]: `movies.iloc[[3,5,6]]`

Out[19]:		title_x	imdb_id	poster_path
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wiki
5	Soni (film)	tt6078866	https://upload.wikimedia.org/wikipedia/en/thum...	https://e
6	Fraud Saiyaan	tt5013008	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w

In [20]: `students.iloc[[0,1,2]]`

Out [20]:

	iq	marks	package
name			
marina	100	80	10
manisha	90	70	7
eshu	120	100	14

selecting both rows and columns

In [21]: `movies.iloc[1:4,1:4]`

Out [21]:

	imdb_id	poster_path	
1	tt9472208	NaN	https://en.wikipedia
2	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/
3	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org

In [22]: `movies.loc[0:3,'imdb_id':'is_adult']`

Out [22]:

	imdb_id	poster_path	
0	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/
1	tt9472208	NaN	https://en.wikipedia
2	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org/
3	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wikipedia.org

filtering dataframe

In [34]: `ipl.sample(5)`

Out [34]:

	ID	City	Date	Season	MatchNumber	Team1	Team2
131	1254060	Chennai	2021-04-11	2021	3	Kolkata Knight Riders	Sunrisers Hyderabad
740	501234	Kolkata	2011-04-30	2011	37	Kolkata Knight Riders	Kings XI Punjab
741	501232	Bangalore	2011-04-29	2011	35	Royal Challengers Bangalore	Pune Warriors
502	734027	Kolkata	2014-05-22	2014	49	Kolkata Knight Riders	Royal Challengers Bangalore
522	733989	Ahmedabad	2014-05-08	2014	30	Rajasthan Royals	Sunrisers Hyderabad

In [29]:

```
# find all the final winners
ipl[ipl['MatchNumber'] == 'Final'][['Season', 'WinningTeam']]
```

Out [29]:

	Season	WinningTeam
0	2022	Gujarat Titans
74	2021	Chennai Super Kings
134	2020/21	Mumbai Indians
194	2019	Mumbai Indians
254	2018	Chennai Super Kings
314	2017	Mumbai Indians
373	2016	Sunrisers Hyderabad
433	2015	Mumbai Indians
492	2014	Kolkata Knight Riders
552	2013	Mumbai Indians
628	2012	Kolkata Knight Riders
702	2011	Chennai Super Kings
775	2009/10	Chennai Super Kings
835	2009	Deccan Chargers
892	2007/08	Rajasthan Royals

```
In [35]: # how many super over finishes have occurred
ipl[ipl['SuperOver'] == 'Y'].shape[0]
```

Out[35]: 14

```
In [36]: # how many matches has csk won in kolkata
ipl[(ipl['City'] == 'Kolkata') & (ipl['WinningTeam'] == 'Chennai Super Ki
```

Out [36]:

	ID	City	Date	Season	MatchNumber	Team1	Team2	Venue
224	1178404	Kolkata	2019-04-14	2019	29	Kolkata Knight Riders	Chennai Super Kings	Eden Gardens
602	598022	Kolkata	2013-04-20	2013	26	Kolkata Knight Riders	Chennai Super Kings	Eden Gardens
641	548368	Kolkata	2012-05-14	2012	63	Kolkata Knight Riders	Chennai Super Kings	Eden Gardens
827	419113	Kolkata	2010-03-16	2009/10	8	Kolkata Knight Riders	Chennai Super Kings	Eden Gardens
908	336025	Kolkata	2008-05-18	2007/08	41	Kolkata Knight Riders	Chennai Super Kings	Eden Gardens

In [39]: `movies.sample(4)`

		title_x	imdb_id	poster_path
Out [39]:	1091	Johnny Gaddaar	tt1077248	https://upload.wikimedia.org/wikipedia/en/thum... https://

1578	Aks (2001 film)	tt0289845	https://upload.wikimedia.org/wikipedia/en/thum... https://
------	-----------------	-----------	--

665	Jodi Breakers	tt2191721	https://upload.wikimedia.org/wikipedia/en/thum... http
-----	---------------	-----------	--

778	Zindagi Na Milegi Dobara	tt1562872	https://upload.wikimedia.org/wikipedia/en/thum... https://en
-----	--------------------------	-----------	--

```
In [46]: # movies with rating higher than 8 and votes>10000
movies[(movies['imdb_rating'] > 8) & (movies['imdb_votes'] > 10000)].shape
```

Out[46]: 43

```
In [52]: # Action movies with rating higher than 7.5
movies1 = movies['genres'].str.contains('Action')
movies2 = movies['imdb_rating'] > 7.5
```

```
In [54]: movies[movies1 & movies2].shape[0]
```

Out[54]: 33

ADDING NEW COLUMNS

```
In [57]: movies['Country'] = 'India'
movies.head(4)
```

Out [57]:

	title_x	imdb_id	poster_path	
0	Uri: The Surgical Strike	tt8291224	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
1	Battalion 609	tt9472208	NaN	https://
2	The Accidental Prime Minister (film)	tt6986710	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
3	Why Cheat India	tt8108208	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.

```
In [60]: ## dropping nan values permentally
movies.dropna(inplace = True)
```

```
In [67]: ## adding new colum called lead actors from actors column
movies['lead_actors'] = movies['actors'].str.split('|').apply(lambda x:x[
```

```
In [68]: movies.head(3)
```

Out [68]:

	title_x	imdb_id	poster_path	
11	Gully Boy	tt2395469	https://upload.wikimedia.org/wikipedia/en/thum...	https://i
34	Yeh Hai India	tt5525846	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.w
37	Article 15 (film)	tt10324144	https://upload.wikimedia.org/wikipedia/en/thum...	https://en.wik

```
In [ ]: ## astype function
```

```
In [69]: # astype
ipl.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                     950 non-null   int64
1   City                   899 non-null   object
2   Date                   950 non-null   object
3   Season                 950 non-null   object
4   MatchNumber            950 non-null   object
5   Team1                  950 non-null   object
6   Team2                  950 non-null   object
7   Venue                  950 non-null   object
8   TossWinner             950 non-null   object
9   TossDecision           950 non-null   object
10  SuperOver              946 non-null   object
11  WinningTeam            946 non-null   object
12  WonBy                  950 non-null   object
13  Margin                 932 non-null   float64
14  method                 19 non-null    object
15  Player_of_Match        946 non-null   object
16  Team1Players            950 non-null   object
17  Team2Players            950 non-null   object
18  Umpire1                950 non-null   object
19  Umpire2                950 non-null   object
dtypes: float64(1), int64(1), object(18)
memory usage: 148.6+ KB
```

```
In [72]: ## changing the dtype of id column
ipl['ID'] = ipl['ID'].astype('int32')
```

```
In [73]: ipl.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   ID                    950 non-null    int32
1   City                  899 non-null    object
2   Date                  950 non-null    object
3   Season                950 non-null    object
4   MatchNumber           950 non-null    object
5   Team1                 950 non-null    object
6   Team2                 950 non-null    object
7   Venue                 950 non-null    object
8   TossWinner            950 non-null    object
9   TossDecision          950 non-null    object
10  SuperOver             946 non-null    object
11  WinningTeam           946 non-null    object
12  WonBy                 950 non-null    object
13  Margin                932 non-null    float64
14  method                19 non-null     object
15  Player_of_Match       946 non-null    object
16  Team1Players           950 non-null    object
17  Team2Players           950 non-null    object
18  Umpire1               950 non-null    object
19  Umpire2               950 non-null    object
dtypes: float64(1), int32(1), object(18)
memory usage: 144.9+ KB
```

```
In [75]: # chaning the dtype of team1 nnd team2
ipl['Team1'] = ipl['Team1'].astype('category')
ipl['Team2'] = ipl['Team2'].astype('category')
```

```
In [76]: ipl.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 950 entries, 0 to 949
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                    950 non-null    int32
1   City                  899 non-null    object
2   Date                  950 non-null    object
3   Season                950 non-null    object
4   MatchNumber           950 non-null    object
5   Team1                 950 non-null    category
6   Team2                 950 non-null    category
7   Venue                 950 non-null    object
8   TossWinner            950 non-null    object
9   TossDecision          950 non-null    object
10  SuperOver             946 non-null    object
11  WinningTeam           946 non-null    object
12  WonBy                 950 non-null    object
13  Margin                932 non-null    float64
14  method                19 non-null     object
15  Player_of_Match       946 non-null    object
16  Team1Players           950 non-null    object
17  Team2Players           950 non-null    object
18  Umpire1               950 non-null    object
19  Umpire2               950 non-null    object
dtypes: category(2), float64(1), int32(1), object(16)
memory usage: 133.2+ KB

```

In []: